

Name: SAMPLE ANSWERS Roll number: _____ House: _____

CS348 Computer Networks MidSem, 17 Sep 2025, 13:30-15:30, MaxMarks:30

General instructions

- Write only in the space provided. Answer briefly but crisply. Write neatly and clearly.
- You are allowed to refer to your own hand-written notes only.
- All answers/calculations have to be (briefly) explained. State any necessary assumptions.

Section A – Short Answer Questions [2 x 4 = 8 marks]

1. **Circuit vs Packet Switched:** Application-A generates constant length frames every 20ms, and requires a guaranteed upper bound on the delay each frame experiences while traversing the network from the sender to the receiver. Application-B also requires a guaranteed upper bound on the delay, but generates frames of different sizes every 20ms (e.g. video conferencing). Application-C generates frames of different sizes every 20ms, but does not require any guaranteed delay bounds.

a) A packet switched network is suited best for which of the above applications, and why?

C - variable data rate (stat. multiplexing), no delay bound

b) A circuit switched network is suited best for which of the above applications, and why?

A - fixed data rate (no stat. multiplexing), delay bounded

2. **Ethernet vs WiFi MAC:** Mention any one similarity between ethernet MAC and WiFi MAC. Also mention any one difference between the two.

Similarities - carrier sense, exponential backoff on collision

Diff - no collision detection in WiFi, has immediate ACK

3. **HDLC:** In HDLC, the delimiter bit-pattern used is 01111110. The sender stuffs a '0' after every 5 consecutive 1's in the data stream. For each of the following bit-patterns received at a HDLC link-layer receiver, in the middle of a frame, (a) remove any stuffed bits, and (b) indicate if there are any errors. In your answer, be sure to have a space between every 4 bits, for readability.

a) 1011 1101 1111 0001 1

↓ ↓ ↓ ↓ ↓ → 0 after 5 1's is unstuffed
1011 1101 1111 0001 1

b) 1100 1111 1000 0010 0

↓ ↓ ↓ ↓ ↓
1100 1111 1000 0010 0

c) 0001 1111 1000 1100 1

→ delimiter inside frame ⇒ error

d) 0110 1111 1110 0110 1

→ 7 1's in sequence ⇒ error

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4. **Supernetting:** An organization has 3 networks as follows: 190.168.20.0/24, 190.168.21.0/25, 190.168.21.128/25. How will the organization appear in the forwarding table of a "far away" router which has the same outgoing interface for all 3 networks? Explain briefly.

$$\begin{array}{l} 190.168.0001\ 0100.0000\ 0000/24 \\ 190.168.0001\ 0101.0000\ 0000/25 \\ 190.168.0001\ 0101.1000\ 0000/25 \end{array} \rightarrow 190.168.20.0/23$$

Section B – In-Depth Questions

5. **WiFi and Sliding Window Protocol:** WiFi link layer (in 802.11 a/b/g) uses the stop-and-wait protocol. And as per the MAC protocol, an ACK is sent *exactly* after SIFS, on successful reception of a data frame.

a) [1 mark] How many bits are required for the sequence number in the data frames? Why?

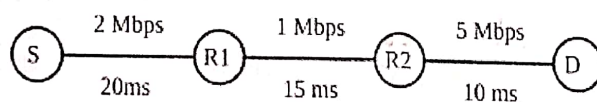
Stop & wait $\Rightarrow$ 1 bit needed for data seq. num

b) [3 marks] How many bits are required for the sequence number in the ACK frames? Why? (actual in WiFi is 12)

Hint: pay attention to how many marks this carries

Unlike regular stop-and-wait where ACK can be for any prev. frame, here NO confusion
 $\Rightarrow$ 0 bits needed = actual in WiFi ACK

6. **Store-and-Forward:** Consider the following store-and-forward datagram switched network between S and D. The link speeds and the propagation delays at each hop are as shown. A packet P1 of 1250 bytes is sent from S to D, followed immediately by a packet P2 of the same size. There is no other traffic on the network. Ignore processing delays.



a) [1 mark] At what time does D finish receiving P1?

$$Tx\ time = 10K\ bits / 2M\ (5ms) + 10ms + 2ms = 17ms$$

$$17ms + prop\ times = 62ms$$

b) [3 marks] At what time does D finish receiving P2? Hint: your answering approach matters

Method matters. Long method: max 1.5 marks

Effective data rate $\hat{=}$ min $\hat{=}$ 1 Mbps

So P2 will be behind P1 by 10ms = 72ms

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7. **IP Forwarding [1x4=4 marks]:** Shown below is a forwarding table at a router in the public Internet (values are in hexadecimal, for easy conversion to binary). Explain your answers below briefly. *Hint: Pay attention that the router is on the public Internet.*

Destination network	Next-hop
0.0.0.0/0	N1
80.0.0.0/2	N2
83.F0.0.0/12	N3
83.FC.0.0/14	N4
83.FD.0.0/16	N5

- a) For a packet with destination address (in hexa) 83.FE.B0.0A, what is the next-hop chosen?

N5 does not match. C & E match in 2 bits. So N4

- b) For a packet with destination address (in hexa) B3.FE.B0.0A, what is the next-hop chosen?

8 & B match in 2 bits. So N2

- c) For a packet with destination address (in hexa) C3.FE.B0.0A, what is the next-hop chosen?

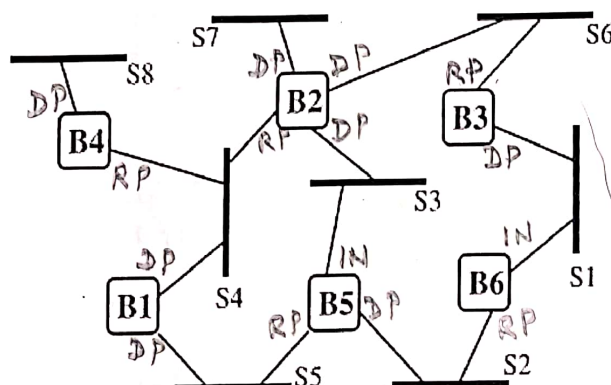
Even N2 does not match, so default N1

- d) For a packet with destination address (in hexa) 0A.B0.FE.83, what is the next-hop chosen?

Trick question 10.0.0.0/8 is private $\Rightarrow$ dropped

8. **Spanning Tree [5 marks]:** Answer the following with respect to an extended ethernet LAN with 6 bridges and 8 LAN segments as shown. Explain your answers briefly.

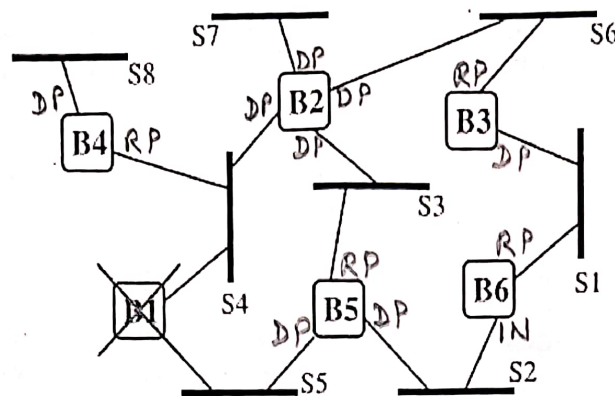
- a) Mark each port of each bridge as RP (root port), DP (designated port), or IN (inactive), after the spanning tree protocol has converged. Mark in pencil and in the diagram itself below.



B1 is root
check 2 cycles
- both broken
due to IN

- b) Repeat the above exercise after bridge B1 fails, and the spanning tree protocol has converged once again. Mark in pencil and in the diagram itself below (next page).

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B2 is root
One cycle broken
by one IN

9. CSMA/CA [5 marks]: Consider 3 WiFi nodes A, B, C in range of each other. The sequence of random numbers generated by each node is as follows: $A=[3,2,5]$, $B=[1,4,0]$, $C=[0,2,3]$. The only frames under consideration are as follows: A has 5 broadcast frames in its queue, B has 2 frames destined to C in its queue. RTS/CTS is turned off by configuration, and there are no channel errors. Ignore any propagation delays. Show the operation of the MAC protocol, with all relevant delays marked, until successful ACK of B's second frame. Answer in terms of a neat timing diagram. Note: it is best to draw in pencil. "Fold" the timeline if necessary.

B's first frame + ACK, A's first frame (no ACK),
A & B collide, B's second frame + ACK broadcast

